WORKED SOLUTION

STEP I 2018, Question 7

Pure • Solving the Cubic

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s1-q7

The key idea

The substitution is a Möbius transformation chosen so that, after clearing denominators, the z^2 and z terms cancel and only z^3 and a constant are left. A cubic of the form $az^3 + b = 0$ is easy to solve, so the whole job is to bring a general reduced cubic $x^3 - 3cx + d = 0$ into the special shape. Doing that means finding two numbers from their sum and product, which is a quadratic, and the condition for it to have distinct real roots is where $d^2 > 4c^3$ comes from.

Solution

Part (i)

Put $x = \dfrac{pz+q}{z+1}$ and multiply the equation through by $(z+1)^3$:

$$(pz+q)^3 - 3pq(pz+q)(z+1)^2 + pq(p+q)(z+1)^3 = 0.$$

We need $(pz+q)(z+1)^2 = pz^3 + (2p+q)z^2 + (p+2q)z + q$. Now collect powers of z.

$$\begin{aligned}
z^3: &\quad p^3 - 3p^2q + p^2q + pq^2 = p(p^2 - 2pq + q^2) = p(p-q)^2,\\
z^2: &\quad 3p^2q - 3pq(2p+q) + 3pq(p+q) = 3p^2q - 6p^2q - 3pq^2 + 3p^2q + 3pq^2 = 0,\\
z^1: &\quad 3pq^2 - 3pq(p+2q) + 3pq(p+q) = 3pq^2 - 3p^2q - 6pq^2 + 3p^2q + 3pq^2 = 0,\\
z^0: &\quad q^3 - 3pq^2 + p^2q + pq^2 = q(q^2 - 2pq + p^2) = q(p-q)^2.
\end{aligned}$$

So the equation reduces to $az^3 + b = 0$ with

$$a = p(p-q)^2, \qquad b = q(p-q)^2.$$

Since $p \neq q$ we may divide by $(p-q)^2$, giving the simpler form $pz^3 + q = 0$.

The substitution can be inverted, $z = \dfrac{x-q}{p-x}$, for every $x \neq p$, and $x = p$ is a root of the cubic only when $p(p-q)^2 = 0$. Also $z = -1$ never solves $pz^3 + q = 0$, because

that would need $p = q$. So the roots of the cubic correspond exactly to the roots of $pz^3 + q = 0$ (when $p = 0$ the cubic is just $x^3 = 0$).

Part (ii)

Comparing $x^3 - 3cx + d$ with $x^3 - 3pqx + pq(p+q)$, we need

$$pq = c, \qquad pq(p+q) = d, \qquad \text{that is} \qquad pq = c, \quad p+q = \frac{d}{c},$$

where dividing by c is allowed because $c \neq 0$. Two numbers with sum $\frac{d}{c}$ and product c are the roots of

$$t^2 - \frac{d}{c}\,t + c = 0.$$

This quadratic has distinct real roots exactly when its discriminant is positive:

$$\frac{d^2}{c^2} - 4c > 0 \iff d^2 > 4c^3,$$

multiplying by $c^2 > 0$. So when $d^2 > 4c^3$, taking p and q to be the two roots of the quadratic writes the equation in the required form, with p, q real and distinct.

Part (iii)

Here $c = -2$ and $d = -2$, so $d^2 = 4 > -32 = 4c^3$ and part (ii) applies. We need $p + q = \frac{d}{c} = 1$ and $pq = c = -2$, so p, q are the roots of $t^2 - t - 2 = (t-2)(t+1) = 0$. Take $p = 2$, $q = -1$. By part (i) the equation becomes

$$2z^3 - 1 = 0, \qquad z^3 = \tfrac{1}{2},$$

whose only real root is $z = 2^{-1/3}$. Write $u = 2^{1/3}$, so $z = \frac{1}{u}$. Then

$$x = \frac{2z-1}{z+1} = \frac{2-u}{1+u}.$$

Ask yourself Can this be tidied up, given that the denominator involves a cube root?

Yes. Rationalise using $1 + u^3 = (1+u)(1-u+u^2)$, which equals 3 since $u^3 = 2$:

$$x = \frac{(2-u)(1-u+u^2)}{3} = \frac{2-3u+3u^2-u^3}{3} = \frac{3u^2-3u}{3} = u^2 - u.$$

So the real root is

$$x = 2^{2/3} - 2^{1/3} = \sqrt[3]{4} - \sqrt[3]{2}.$$

It is the only real root: $x^3 + 6x - 2$ has derivative $3x^2 + 6 > 0$, so it is strictly increasing. The other two values of z are complex and give the two complex roots. (Choosing $p = -1$, $q = 2$ instead gives $z^3 = 2$ and the same root.)

Part (iv)

Since $x = p$ gives $p^3 - 3p^3 + 2p^3 = 0$, $x - p$ is a factor, and

$$x^3 - 3p^2x + 2p^3 = (x-p)(x^2 + px - 2p^2) = (x-p)^2(x+2p).$$

So the roots are $x = p$ (repeated) and $x = -2p$.

Notice that this is the form of part (i) with $q = p$, which is exactly the case part (ii) left out. Now take $d^2 = 4c^3$ with $c, d \neq 0$. Then $c^3 = \frac{1}{4}d^2 > 0$, so $c > 0$. We look for p with $p^2 = c$ and $2p^3 = d$. Dividing the second by the first suggests

$$p = \frac{d}{2c},$$

and this works: $p^2 = \dfrac{d^2}{4c^2} = \dfrac{4c^3}{4c^2} = c$, and then $2p^3 = 2p \cdot p^2 = 2 \cdot \dfrac{d}{2c} \cdot c = d$. So $x^3 - 3cx + d = x^3 - 3p^2x + 2p^3$, and its roots are

$$x = \frac{d}{2c} \text{ (twice)} \qquad \text{and} \qquad x = -\frac{d}{c}.$$

Checks

- Part (iii): $\sqrt[3]{4} - \sqrt[3]{2} \approx 0.32748$, and $x^3 + 6x - 2$ at this value is 0 to fifteen decimal places. Cardano's formula gives the same root, $\sqrt[3]{1+3} + \sqrt[3]{1-3}$.
- Part (iv): with $c = 1$, $d = 2$ the cubic is $x^3 - 3x + 2 = (x-1)^2(x+2)$, which matches the double root $\frac{d}{2c} = 1$ and the simple root $-\frac{d}{c} = -2$.
- Part (i): with $p = 1$, $q = 0$ the cubic is $x^3 = 0$ and the substitution gives $x = \frac{z}{z+1}$, so $x^3(z+1)^3 = z^3$, which matches $a = 1$, $b = 0$.

Common slips

- Forgetting that c may be negative in part (ii): multiplying the inequality by c^2 is safe, but multiplying by c is not.
- In part (iii), giving all three cube roots of $\frac{1}{2}$ as real roots. Only $z = 2^{-1/3}$ is real.
- In part (iv), taking $p = \sqrt{c}$. That has the wrong sign when $d < 0$; $p = \frac{d}{2c}$ gets the sign right automatically.

Beyond STEP

This is a neat route to Cardano's formula for the cubic. The quadratic $t^2 - \frac{d}{c}t + c = 0$ plays the part of the "resolvent", and the sign of $d^2 - 4c^3$ is (up to a factor) the discriminant of the cubic, which decides whether it has one real root or three. Why cubics and quartics can be solved by radicals while general quintics cannot is the subject

of Galois theory. Allenby's *Rings, Fields and Groups* is a good place to start reading about it.

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